

Dual 555 Drone

Build two independently adjustable square-wave oscillators and mix them into one line-level output.

EVIDENCE STATUS

Calculated circuit relationships + browser-simulated audio. Construction layout is diagram-audited but NOT physically tested.

1. Project purpose

Build two independently adjustable square-wave oscillators and mix them into one line-level output.

How it works

Each timer repeatedly charges and discharges a capacitor. Increasing timing resistance or capacitance increases charge time and lowers pitch. Two nearby frequencies create audible beating.

Calculated relationship

Formula: $t_{HIGH} = 0.693(RA + RB)C$; $t_{LOW} = 0.693(RB)C$; $f = 1/(t_{HIGH} + t_{LOW})$

Substitution: $RA = 10\text{ k}\Omega$, $RB = 22\text{ k}\Omega$, $C = 10\text{ nF}$ -> $t_{HIGH} = 0.222\text{ ms}$, $t_{LOW} = 0.152\text{ ms}$

Result: Calculated frequency: 2672 Hz; duty cycle: 59.3%. Browser estimate: about 2667 Hz.

Safety boundary

Use only a regulated, current-limited 9 V DC supply. Disconnect power before moving parts. Check IC orientation, diode bands, electrolytic polarity, and rail continuity. Connect the output only to a powered speaker, amplifier, or audio-interface input - never directly to a bare low-impedance speaker.

2. Required components

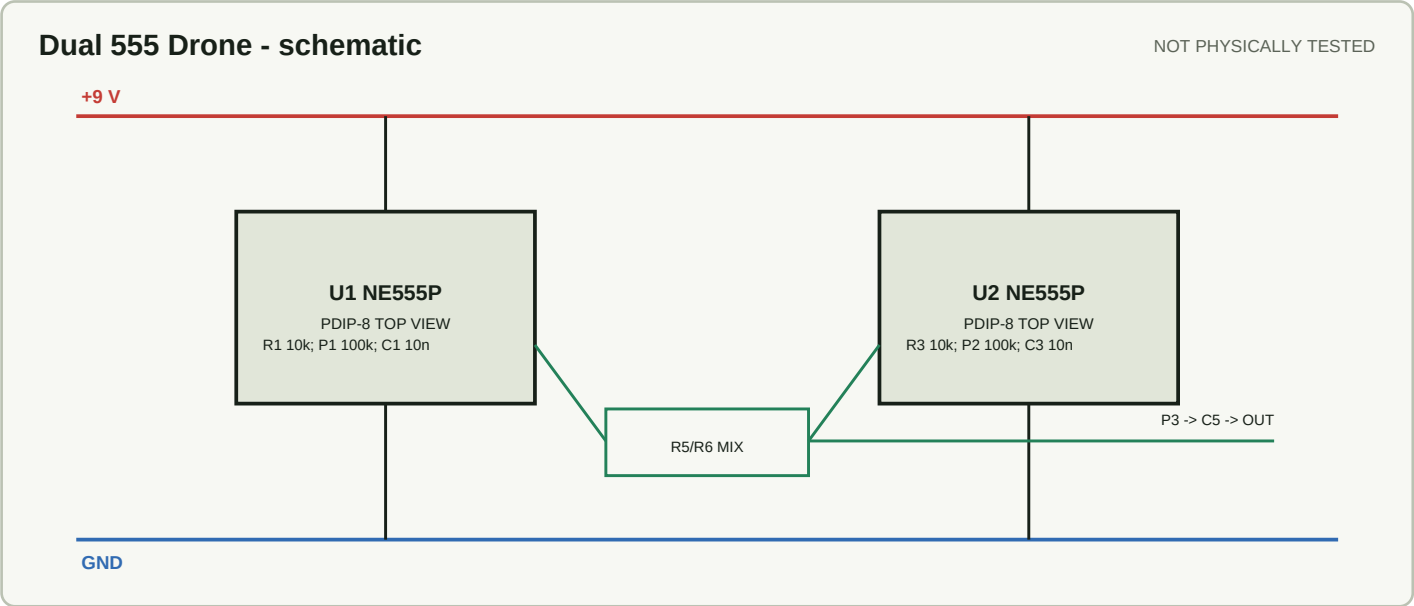
Ref	Exact component / value	Qty
BB1	BB830-style solderless breadboard	1
U1, U2	TI NE555P PDIP-8	2
R1, R3	10 kOhm 1/4 W metal-film	2
P1, P2	Bourns PTV09 100 kOhm linear	2
C1, C3	10 nF film timing capacitor	2
C2, C4	10 nF ceramic capacitor	2
R5, R6	10 kOhm 1/4 W mixing resistor	2
P3	Bourns PTV09 100 kOhm linear	1
C5	1 uF 25 V electrolytic	1
C6	100 nF ceramic bypass	1
J1	6.35 mm mono output jack	1
PS1	Regulated current-limited 9 V DC supply	1

Pinout verification

NE555P package view: TOP VIEW. In the supplied horizontal breadboard layout the notch faces left; DIP numbering proceeds counterclockwise.

Pin	Name	Role
1	GND	Ground
2	TRIGGER	Timing trigger
3	OUTPUT	Timer output
4	RESET	Active-low reset
5	CONTROL	Threshold control
6	THRESHOLD	Timing threshold
7	DISCHARGE	Timing discharge
8	VCC	Positive supply

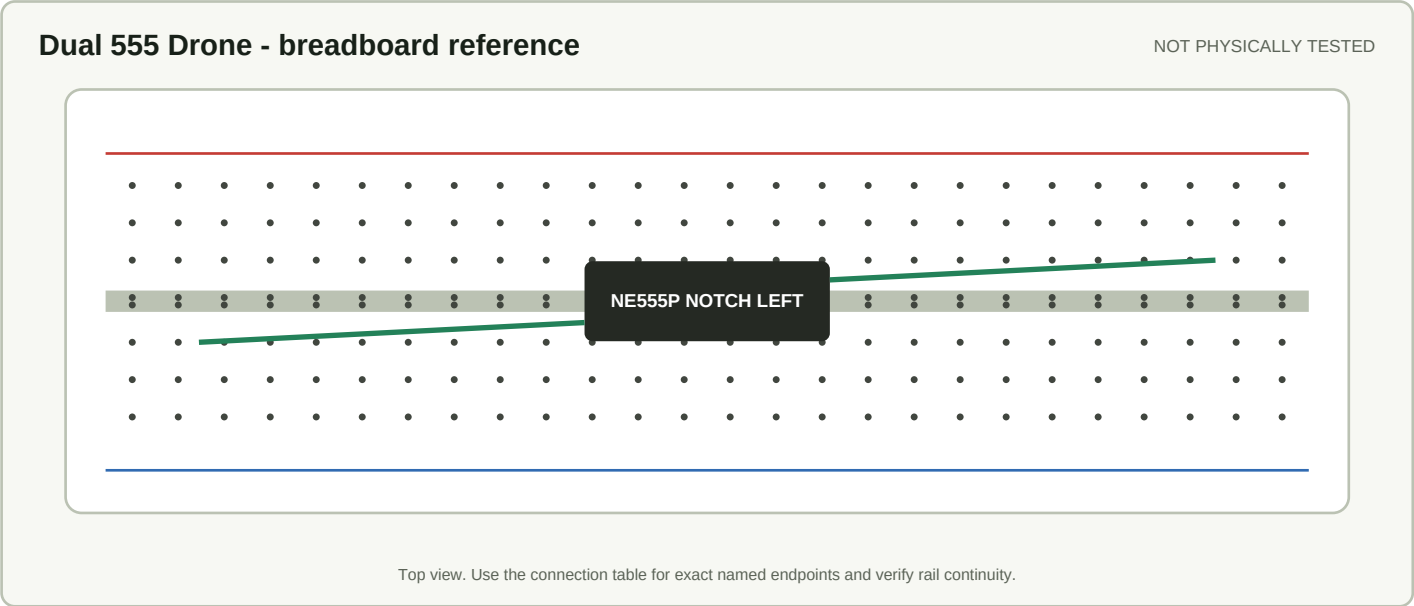
3. Labeled schematic



4. Canonical connections

Net	From	To	Wire
+9V	+ rail	U1 pin 8 and pin 4; U2 pin 8 and pin 4	red
GND	- rail	U1 pin 1; U2 pin 1; J1 sleeve	black
A timing	U1 pin 2	U1 pin 6; C1 to ground	blue
A charge	+9 V through R1	U1 pin 7; P1 from pin 7 to pins 2/6	red/blue
A control	U1 pin 5	C2 to ground	black
Voice A	U1 pin 3	R5 to MIX	yellow
B timing	U2 pin 2	U2 pin 6; C3 to ground	blue
B charge	+9 V through R3	U2 pin 7; P2 from pin 7 to pins 2/6	red/blue
B control	U2 pin 5	C4 to ground	black
Voice B	U2 pin 3	R6 to MIX	orange
Output	R5/R6 junction	P3 lug 3; wiper through C5 to J1 tip	green
Bypass	+9 V	C6 to ground close to U1/U2	black

5. Breadboard reference



Breadboard orientation

This drawing is a top-view placement reference. The connection table is authoritative for every electrical net. Typical breadboards connect A-E in each numbered row and F-J separately across the center trench; power-rail continuity varies by product and must be measured with power removed.

6. Power-off assembly

- 01** With power disconnected, map the breadboard rails using continuity mode.
- 02** Place U1 across the center trench at columns 10-13 and U2 at columns 20-23. Both notches face left.
- 03** Connect each pin 1 to ground, each pin 8 to +9 V, and each pin 4 to +9 V.
- 04** Build U1 timing: join pins 2 and 6; add R1 from +9 V to pin 7; add P1 from pin 7 to pins 2/6; add C1 from pins 2/6 to ground.
- 05** Repeat the timing network for U2 using R3, P2, and C3.
- 06** Connect C2 and C4 from each pin 5 to ground. Put C6 across the rails near the ICs.
- 07** Connect U1 pin 3 through R5 and U2 pin 3 through R6. Join only the far resistor ends at MIX.
- 08** Connect MIX to P3 lug 3, lug 1 to ground, and the wiper through C5 to J1 tip. C5 positive faces the wiper.
- 09** Set P3 low, inspect every connection, then apply current-limited 9 V.

7. Test procedure and expected results

Start with the supply current limit low and the output level down. Use a multimeter in DC-voltage mode with the black probe on the ground rail. Confirm the power and bias readings before connecting audio. Expected readings below are predictions, not measurements.

Checkpoint	Calculated / expected observation
1	U1/U2 pin 8 to pin 1: about 9 V DC
2	Pins 2/6: repeating ramp between about 3 V and 6 V
3	Pins 3: pulse wave; frequency changes with P1/P2
4	Output: mixed pulse tones into a powered audio input

8. Troubleshooting

1. Remove power. Confirm +9 V and ground are not shorted using resistance or continuity mode.
2. Confirm the IC notch direction and locate pin 1 again. A mirrored package is the most serious common wiring error.
3. Check every polarized capacitor and every diode band against the connection table.
4. Measure the supply at the IC pins. If it is missing, repair power wiring before testing signals.
5. Compare one net at a time with the canonical table. Do not change several wires at once.
6. If DC checks pass but sound is missing, trace from output toward input with a powered audio probe or oscilloscope under instructor supervision.
7. Component tolerances change pitch, cutoff, gain, and bias. A difference from the browser model is not automatically a fault.

Sources

TI NE555 datasheet: <https://www.ti.com/lit/ds/symlink/ne555.pdf>

Bourns PTV09 datasheet: <https://www.bourns.com/docs/product-datasheets/PTV09.pdf>

Wire audit statement

The connection table, schematic labels, and breadboard IC orientation were reconciled for this guide. Off-board potentiometers and jacks are identified by terminal name in the table. No physical build or measurement was performed; complete a continuity and power-rail audit before first power-up.